

Companion to Bird & de Moor, Algebra of Programming

§8. Thinning Algorithms

§8.1. Thinning

(8.1)

For $Q : A \rightarrow A$, $\text{thin}(Q) \triangleq (\exists/\exists) \cap (\epsilon \setminus (Q^\circ \epsilon)) : PA \rightarrow PA$ (8.1).

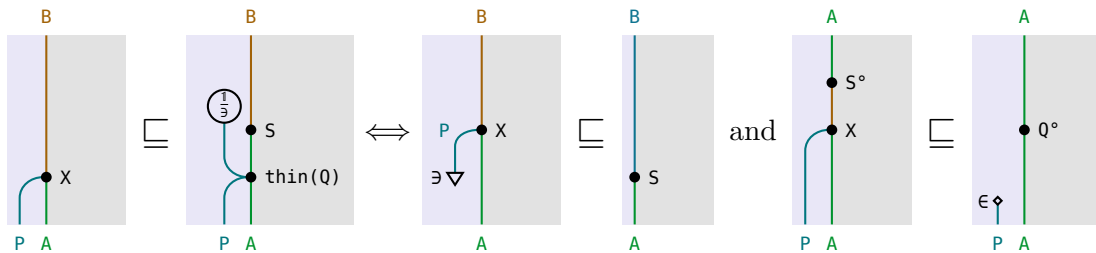
$ys \text{ thin}(Q) xs \iff xs \subseteq ys \wedge (\forall a \in ys. \exists b \in xs. b \ Q \ a)$ the same $^\circ$ as $\triangleq \text{est}$: ϵX runs $a \rightarrow ys \rightarrow xs$, member of ys first, and $Q^\circ \epsilon$ runs $a \rightarrow b \rightarrow xs$, so $(a, b) \in Q^\circ$ reads $b \ Q \ a$; at $Q^\circ \leq$ every $a \in ys$ keeps some $b \leq a$ in xs , the end $\text{est}(\leq)$ picks

(8.1a)

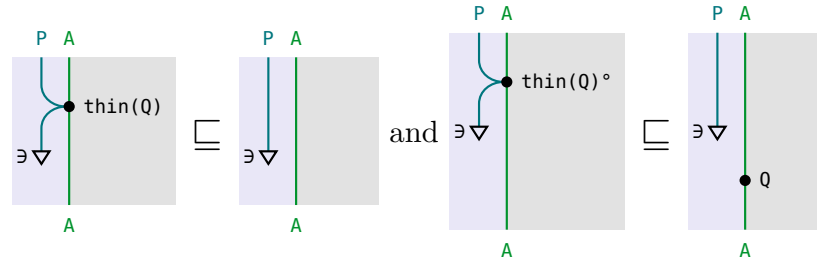
the law	what it says
$X \in \frac{S}{\exists} \text{ thin}(Q) \iff X \exists \in S$ and $S^\circ X \in Q^\circ \epsilon$	everything kept is an S -value, and every S -value has a Q -lower bound among the kept ones
$\text{thin}(Q) \exists \in \exists$	everything a thinning keeps was in the set
$\text{thin}(Q)^\circ \exists \in \exists \ Q$	every element of the set has a Q -lower bound among the kept ones
$Q \subseteq R \implies \text{thin}(Q) \subseteq \text{thin}(R)$	the fewer pairs Q relates, the fewer subsets count as thinnings
$1 \subseteq Q \implies 1 \subseteq \text{thin}(Q)$ $Q \ Q \subseteq Q \implies \text{thin}(Q) \ \text{thin}(Q) \subseteq \text{thin}(Q)$	keeping everything is always a legal thinning
$Q \subseteq R \wedge 1 \subseteq Q \wedge R \ R \subseteq R \implies \text{est}(R) = \text{thin}(Q) \ \text{est}(R)$ $Q \subseteq R, 1 \subseteq Q, R R \subseteq R$ — (8.1.2a); weaker than the book's "both preorders": Q transitive is never used and $1 \subseteq R$ follows	thin-introduction : thinning first cannot lose an R -minimum
$\text{thin}(Q) \supseteq \text{est}(Q) \ \frac{1}{\exists}$ (8.2) — (8.2)	thin-elimination : keeping one element is a thinning, but its domain is the sets $\text{est}(Q)$ is defined on
$\frac{S}{\exists} \text{ thin}(Q) \supseteq \frac{S}{\exists} \text{est}(R) \ \frac{1}{\exists}$ (8.3), $R \cap (S^\circ S) \subseteq Q$ — thin-elimination variant	the usable variant: R need only refine Q between values S gives one argument
$P(\text{thin}(Q)) \text{ union} \subseteq \text{union thin}(Q)$ (8.4)	thinning each member set is a thinning of the union

§8.1.1. $X \in (S^\circ \exists) \text{ thin}(Q) \iff X \exists \in S$ and $S^\circ X \in Q^\circ \epsilon$

(8.1.1a)



(8.1.1b)

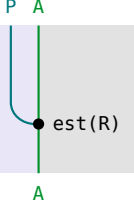
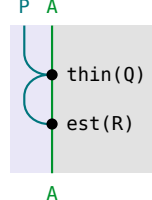


(8.1.1a) at $X \triangleq \text{thin}(Q)$, $S \triangleq \exists$: the left side is $\text{thin}(Q) \subseteq \text{thin}(Q)$, since $\exists \exists = \wedge(\exists) = 1$

§8.1.2. $\text{est}(R) = \text{thin}(Q) \ \text{est}(R)$ given $Q \subseteq R, 1 \subseteq Q, R R \subseteq R$

$\text{est}(R) \sqsubseteq \text{thin}(Q) \text{ est}(R)$ (8.1.2a)

the $\sqsubseteq$ half: keeping everything is a thinning — $1 \sqsubseteq Q$

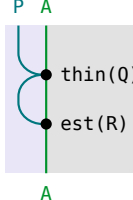
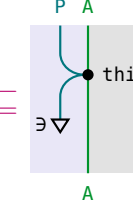
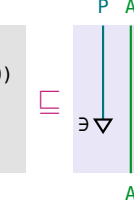



(a) (b)

(a)	$\text{est}(R)$	$1 \sqsubseteq \text{thin}(Q)$ — (8.1a)
(b)	$\sqsubseteq \text{thin}(Q) \text{ est}(R)$	

$\text{thin}(Q) \text{ est}(R) \sqsubseteq \exists$ (8.1.2b)

the $\sqsubseteq$ half, first condition of the UP of est at $X \triangleq \text{thin}(Q) \text{ est}(R)$ — est-up

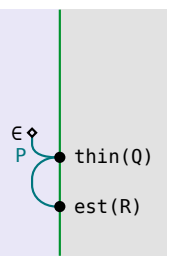
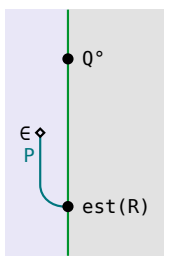
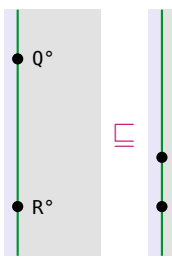
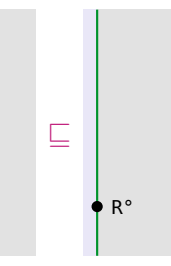




(a) (b) (c)

(a)	$\text{thin}(Q) \text{ est}(R)$	$\text{est}(R) \sqsubseteq \exists$ — est-laws
(b)	$\sqsubseteq \text{thin}(Q) \exists$	$\text{thin}(Q) \exists \sqsubseteq \exists$ — (8.1.1a)
(c)	$\sqsubseteq \exists$	

$\in \text{thin}(Q) \text{ est}(R) \sqsubseteq R^\circ$ (8.1.2c)

the $\sqsubseteq$ half, second condition — $Q \sqsubseteq R$, R transitive


(a) (b) (c) (d) (e)

(a)	$\in \text{thin}(Q) \text{ est}(R)$	$\in \text{thin}(Q) \sqsubseteq Q^\circ \in$ — (8.1.1a)
(b)	$\sqsubseteq Q^\circ \in \text{est}(R)$	$\in \text{est}(R) \sqsubseteq R^\circ$ — est-laws
(c)	$\sqsubseteq Q^\circ R^\circ$	$Q \sqsubseteq R$
(d)	$\sqsubseteq R^\circ R^\circ$	R transitive
(e)	$\sqsubseteq R^\circ$	

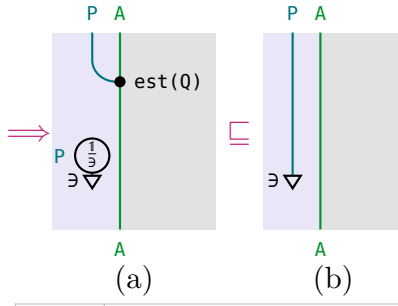
$$\text{est}(Q) \stackrel{1}{\exists} \sqsubseteq \text{thin}(Q)$$

(8.2)

the singleton holding a Q -least member of a set is a thinning of that set

$$\text{est}(Q) \stackrel{1}{\exists} \ni \sqsubseteq \ni$$

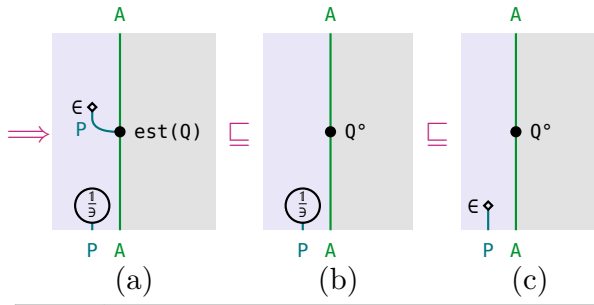
every member of the singleton is a member of the set



(a)	$\Rightarrow \text{est}(Q) \stackrel{1}{\exists} \ni$	$\stackrel{1}{\exists} \ni = 1$ — pow-laws; $\text{est}(Q) \sqsubseteq \ni$ — est-laws
(b)	$\sqsubseteq \ni$	$\triangleq \text{thin}, \epsilon \setminus (Q^\circ \epsilon)$ half

$$\epsilon \text{ est}(Q) \stackrel{1}{\exists} \sqsubseteq Q^\circ \epsilon$$

the singleton's member stands in Q to every member of the set



(a)	$\Rightarrow \epsilon \text{ est}(Q) \stackrel{1}{\exists}$	$\epsilon \text{ est}(Q) \sqsubseteq Q^\circ$ — est-up
(b)	$\sqsubseteq Q^\circ \stackrel{1}{\exists}$	$\stackrel{1}{\exists} \sqsubseteq \epsilon$ — pow-laws
(c)	$\sqsubseteq Q^\circ \epsilon$	

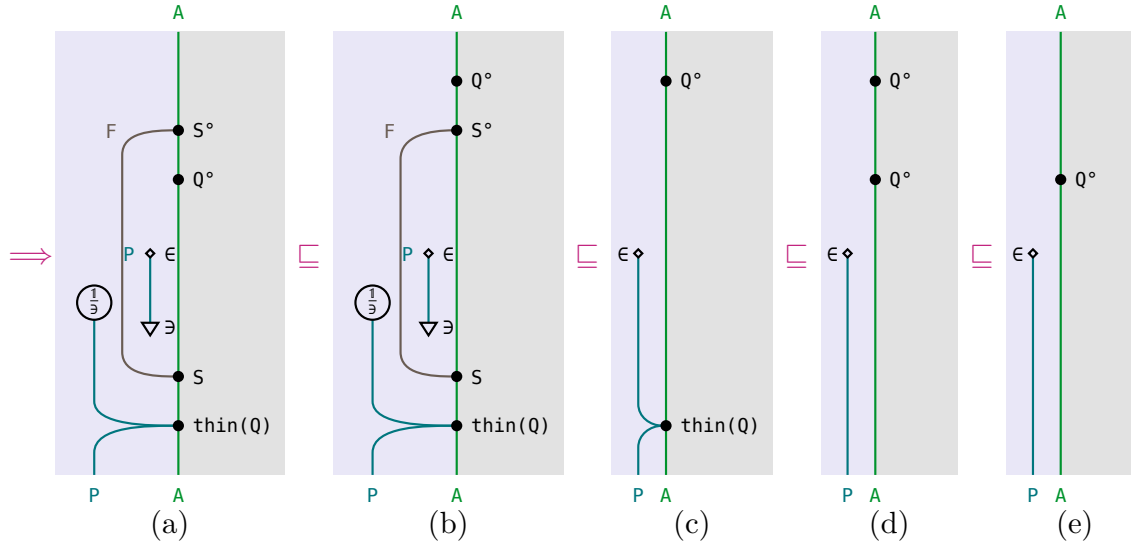
$$Q \subseteq Q \wedge \text{MonoAlg}(S, Q) \Rightarrow (\frac{F(\exists) S}{\exists} \text{thin}(Q)) \subseteq \frac{(S)}{\exists} \text{thin}(Q)$$

Theorem 8.1

thinning at every step of the reduce is a thinning of the whole candidate set — S monotonic on Q , Q a preorder

$$(\frac{F(\exists) S}{\exists} \text{thin}(Q)) \supseteq (S) \text{ and } (S) \circ (\frac{F(\exists) S}{\exists} \text{thin}(Q)) \subseteq Q^\circ \in$$

(8.1a) at $X \triangleq (\frac{F(\exists) S}{\exists} \text{thin}(Q))$, (S) for its S

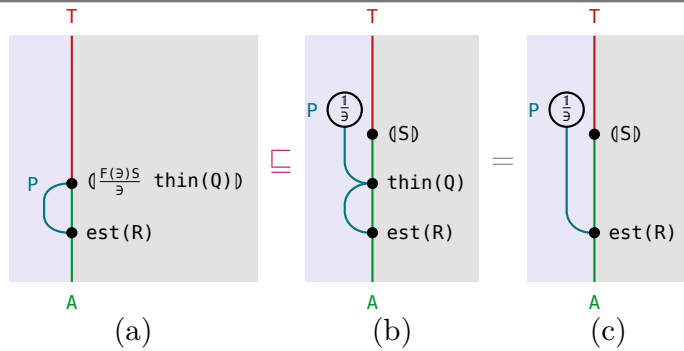


(a)	$\Rightarrow S^\circ F(Q^\circ \in) \frac{F(\exists) S}{\exists} \text{thin}(Q)$	$S^\circ F(Q^\circ) \subseteq Q^\circ S^\circ$ — mon-str, relator-laws
(b)	$\sqsubseteq Q^\circ S^\circ F(\in) \frac{F(\exists) S}{\exists} \text{thin}(Q)$	$S^\circ F(\in) \frac{F(\exists) S}{\exists} \subseteq \in$ — pow-laws
(c)	$\sqsubseteq Q^\circ \in \text{thin}(Q)$	$\in \text{thin}(Q) \subseteq Q^\circ \in$ — $\triangleq$ thin, adj-all
(d)	$\sqsubseteq Q^\circ Q^\circ \in$	Q a preorder
(e)	$\sqsubseteq Q^\circ \in$	

$$Q \subseteq R \wedge 1 \subseteq Q \wedge Q \subseteq Q \wedge R \subseteq R \wedge \text{MonoAlg}(S, Q) \Rightarrow (\frac{F(\exists) S}{\exists} \text{thin}(Q)) \text{ est}(R) \subseteq \frac{(S)}{\exists} \text{est}(R)$$

Corollary 8.1

the thinning fold refines the optimisation problem itself — S monotonic on Q , $Q \subseteq R$, both preorders



(a)	$(\frac{F(\exists) S}{\exists} \text{thin}(Q)) \text{ est}(R)$	thinning theorem
(b)	$\sqsubseteq \frac{(S)}{\exists} \text{thin}(Q) \text{ est}(R)$	$\text{est}(R) = \text{thin}(Q) \text{ est}(R)$ — (8.1a)
(c)	$= \frac{(S)}{\exists} \text{est}(R)$	

§8.2. Paths in a layered network

(8.2a)

$F(A, X) = A + A \times X$, $L = \text{list}^+$ with initial algebra $\alpha \triangleq [\text{wrap}, \text{cons}] : F(A, LA) \rightarrow LA$.

$\text{wrapz} \triangleq \langle \text{wrap}, \text{zero} \rangle$, $\text{consw}(a, (xs, n)) = ([a] \# xs, \text{wt}(a, \text{head}(xs)) + n)$.

$\text{cost} \triangleq ([\text{wrapz}, \text{consw}]) \pi_2$, $([\text{wrapz}, \text{consw}]) = \langle 1, \text{cost} \rangle$, $R \triangleq \text{cost} \leq \text{cost}^\circ$.

$Q \triangleq R \cap (\text{head head}^\circ)$, $S \triangleq F(1, \exists) \alpha$, $\frac{F(\exists, 1)}{\exists} = [1 \text{ P}(l), \text{cpl } \text{P}(r)]$, $\frac{F(1, \exists)}{\exists} = [\frac{1}{\exists} \text{ P}(l), \text{cpr } \text{P}(r)]$, $\text{step} \triangleq \text{cpr } \text{P}(\text{cons}) \text{ est}(R)$.

(8.2b)

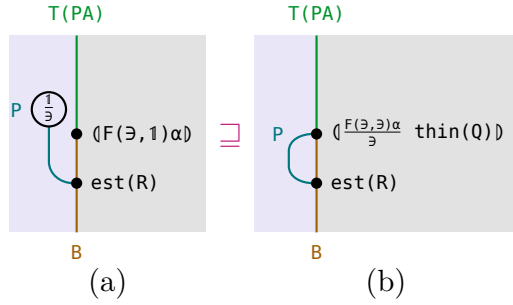
the law	what it says
$F(\exists, Q) \alpha \subseteq F(\exists, 1) \alpha Q$	$F(\exists, 1) \alpha$ is monotonic on Q ; on R it is not, since the next edge can cost arbitrarily much
$S \text{ head} \subseteq [1, \pi_1] \quad S \triangleq F(1, \exists) \alpha$	$S \text{ head}$ is simple, which gives $R \cap (S^\circ S) \subseteq Q$: between two paths S builds from one argument, equal cost and equal head already means Q

$$Q \sqsubseteq R \wedge 1 \sqsubseteq Q \wedge Q \sqsubseteq Q \wedge R^\circ \sqsubseteq R^\circ \wedge \text{MonoAlg}(F(\exists, 1) \alpha, Q) \wedge R \sqcap ((F(1, \exists) \alpha)^\circ F(1, \exists) \alpha) \sqsubseteq Q \quad (8.2c)$$

$$\Rightarrow \langle \frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists)}{\exists} \alpha \text{ est}(R)) \rangle \text{ est}(R) \sqsubseteq \langle \frac{F(\exists, 1)}{\exists} \alpha \rangle \text{ est}(R)$$

a least-cost path in a layered network, as a fold over the layers

$$= \frac{L(\exists)}{\exists} \text{ est}(R)$$



$$\begin{aligned} (a) & \quad \langle \frac{F(\exists, 1)}{\exists} \alpha \rangle \text{ est}(R) \\ (b) & \quad \sqsubseteq \langle \frac{F(\exists, \exists)}{\exists} \alpha \text{ thin}(Q) \rangle \text{ est}(R) \end{aligned}$$

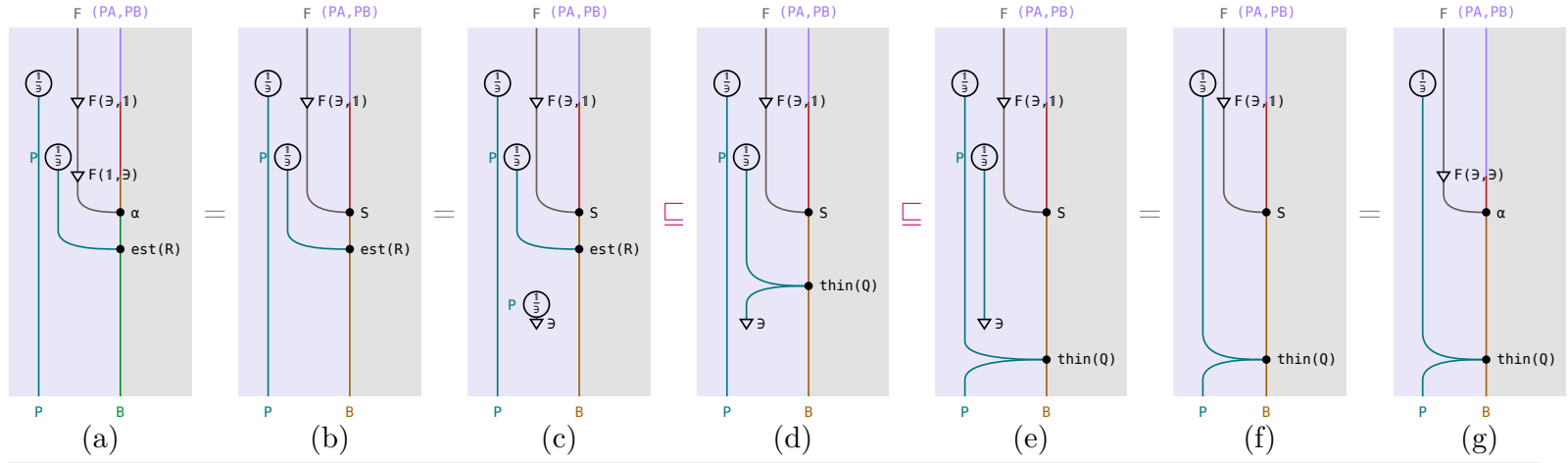
Corollary 8.1, (8.2b)

$\sqsubseteq$ (8.2d) under $\langle \rangle$ monotonic: the whole chain runs inside the reduce, and the $\text{est}(R)$ behind it never moves.

$$R \sqcap (S^\circ S) \sqsubseteq Q \Rightarrow \frac{F(\exists, 1)}{\exists} P(\frac{S}{\exists} \text{ est}(R)) \sqsubseteq \frac{F(\exists, \exists)}{\exists} \alpha \text{ thin}(Q) \quad (8.2d)$$

thinning the algebra of a layered network costs no more than taking the program's two cases

the algebra of (8.2c) row 2



(a)	$\frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists)}{\exists} P(\alpha) \text{ est}(R))$	$\frac{F(1, \exists)}{\exists} P(\alpha) = \frac{S}{\exists}$
(b)	$= \frac{F(\exists, 1)}{\exists} P(\frac{S}{\exists} \text{ est}(R))$	$1 = P(\frac{1}{\exists}) E(\exists)$
(c)	$= \frac{F(\exists, 1)}{\exists} P(\frac{S}{\exists} \text{ est}(R) \frac{1}{\exists}) E(\exists)$	$\frac{S}{\exists} \text{ est}(R) \frac{1}{\exists} \sqsubseteq \frac{S}{\exists} \text{ thin}(Q)$ if $R \sqcap (S^\circ S) \sqsubseteq Q$ — (8.1a), (8.2b)
(d)	$\sqsubseteq \frac{F(\exists, 1)}{\exists} P(\frac{S}{\exists} \text{ thin}(Q)) E(\exists)$	$P(\text{thin}(Q)) E(\exists) \sqsubseteq E(\exists) \text{ thin}(Q)$ — (8.1a)
(e)	$\sqsubseteq \frac{F(\exists, 1)}{\exists} P(\frac{S}{\exists}) E(\exists) \text{ thin}(Q)$	$P(\frac{S}{\exists}) E(\exists) = E(S), \frac{R}{\exists} E(S) = \frac{RS}{\exists}$ — pow-laws
(f)	$= \frac{F(\exists, 1)}{\exists} \frac{S}{\exists} \text{ thin}(Q)$	$F(\exists, 1)F(1, \exists) = F(\exists, \exists)$ $\frac{F(\exists, \exists)}{\exists} \alpha = \{\text{raze } x, /: \backslash : y\}$
(g)	$= \frac{F(\exists, \exists)}{\exists} \alpha \text{ thin}(Q)$	

$$= \text{cp}(F) P(\alpha) \text{ est}(R) = [\text{wrap}, \text{step}]$$

at the layered network $(F(A, X) = A + A \times X, B \text{ the paths } V^+, \alpha = [\text{wrap}, \text{cons}]): \frac{F(1, \exists)}{\exists} P(\alpha) \text{ est}(R) = [\text{wrap}, \text{step}] \triangleq \text{cost}$

$$= \frac{F(\exists, 1)}{\exists} = 1 + \text{cpl} \triangleq \text{cost}$$

$$\begin{aligned} F(A, X) &= A + A \times X \\ &\quad \downarrow \quad \downarrow \quad \downarrow \\ F(A', X') &= A' + A' \times X' \end{aligned}$$

(8.2e)

u $F(R, S)$ v holds iff one of:

- u, v are both inl and $d \ R \ d'$: the leaf moves by R;
- u, v are both inr and $p_1 \ R \ q_1 \wedge p_2 \ S \ q_2$: the vertex moves by R, the rest by S.
- An inl never relates to an inr: $R + S \triangleq [R \text{ inl}, S \text{ inr}]$ returns to the summand it came from, and $F(1) = 1$ allows no crossing.
- $F(\exists, \exists)$ relates $\text{inr}(\{a, b\}, \{p, q\})$ to $\text{inr}(a, p), \text{inr}(a, q), \text{inr}(b, p), \text{inr}(b, q)$, and $\text{inl}(\{a, b\})$ to $\text{inl}(a), \text{inl}(b)$.

§8.3. Implementing thin

(8.3a)

$\text{setify} : [A] \rightarrow PA$, $\text{cup} : PA \times PA \rightarrow PA$, $\text{cp}(F) \triangleq \text{subseq}$, $\text{listcp} : F(L) \rightarrow LF$, $\text{sort}(\leq) \triangleq \text{setify}^\circ \text{ ordered}(\leq)$ for $\leq$ a connected preorder.

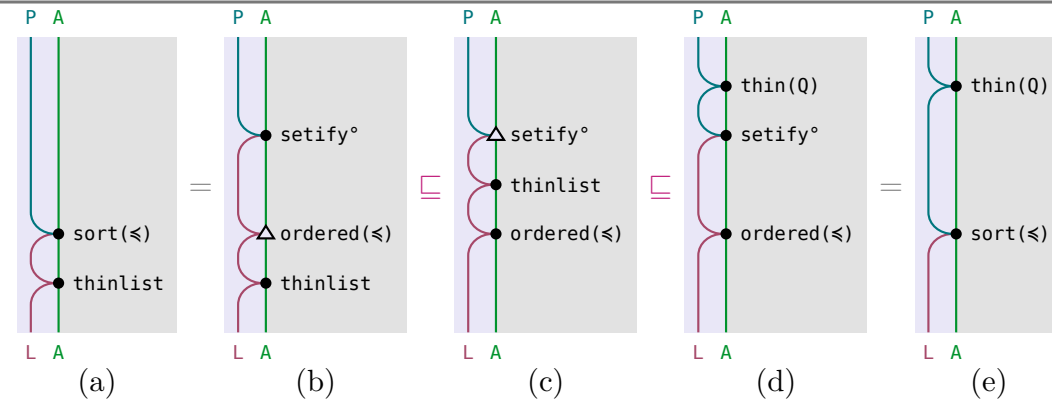
$\text{thinlist}(Q)$ is any $\text{thinlist}(Q) \sqsubseteq \text{subseq}$ with $\text{thinlist}(Q) \text{ setify} \sqsubseteq \text{setify thin}(Q)$; one is $Q[\text{nil}, \text{bump } Q]$, $\text{bump } Q (a, []) = [a]$, $\text{bump } Q (a, [b] \# xs) = (b \text{ } Q \text{ } a \rightarrow [a] \# xs, a \text{ } Q \text{ } b \rightarrow [b] \# xs, [a] \# [b] \# xs)$.

Binary thinning data: $S = (f_1 p_1) \cup (f_2 p_2)$ with p_1, p_2 coreflexive; Q a preorder with $Q \sqsubseteq R$ and both $f_1 p_1, f_2 p_2$ monotonic on Q ; $\leq$ a connected preorder with both f_1, f_2 monotonic on $\leq$; $g_i \triangleq \text{list}(f_i) \text{ filter}(p_i)$.

(8.3b)

the law	what it says
$\text{thinlist}(Q) \text{ } xs = [\text{minlist}(Q) \text{ } xs]$ (8.5), Q connected, xs non-empty	what thinning should come to when it can: one element
$\text{sort}(\leq) \text{ thinlist}(Q) \sqsubseteq \text{thin}(Q) \text{ sort}(\leq)$ (8.6) — (8.6)	thinning a sorted list is a thinning of the set — this is what $\text{thinlist}(Q) \sqsubseteq \text{subseq}$ buys
$\text{sort}(\leq) \text{ minlist}(Q) \sqsubseteq \text{est}(Q)$ (8.7)	a minimum of the sorted list is a minimum of the set
$\text{sort}(f \leq f^\circ) \text{ list}(f) \sqsubseteq P(f) \text{ sort}(\leq)$ (8.8)	shunt a function through a sort
$\text{sort}(\leq) \text{ filter}(p) \sqsubseteq E(p) \text{ sort}(\leq)$ (8.9), p coreflexive	filtering a sorted list sorts the restricted set
$(\text{sort}(\leq) \times \text{sort}(\leq)) \text{ merge}(\leq) \sqsubseteq \text{cup sort}(\leq)$ (8.10)	merging two sorted lists sorts their union
$F(\text{sort}(\leq)) \text{ listcp} \sqsubseteq \text{cp}(F) \text{ sort}(F(\leq))$ (8.11), F linear	listcp is the list implementation of the cartesian product $\text{cp}(F)$

Map(setify) $\wedge$ Coreflexive(ordered($\leq$)) $\wedge$ thinlist $\sqsubseteq$ subseq $\wedge$ ordered($\leq$) subseq $\sqsubseteq$ subseq (8.6)
ordered($\leq$) $\wedge$ thinlist setify $\sqsubseteq$ setify thin(Q) $\Rightarrow$ sort($\leq$) thinlist $\sqsubseteq$ thin(Q) sort($\leq$)
 a thinning of the sorted list lists a thinning of the set — $\leq$ a connected preorder, $\text{thinlist}(Q) \sqsubseteq \text{subseq}$.



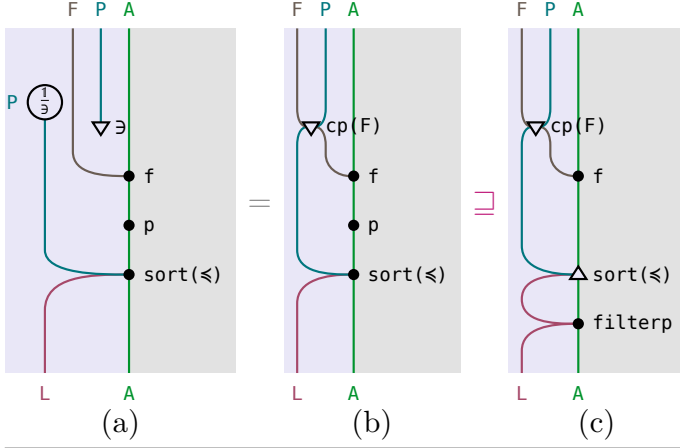
(a)	$\text{sort}(\leq) \text{ thinlist}$	$\text{sort}(\leq) \triangleq \text{setify}^\circ \text{ ordered}(\leq) \text{ — } \triangleq \text{thinlist}$
(b)	$= \text{setify}^\circ \text{ ordered}(\leq) \text{ thinlist}$	$\text{ordered}(\leq) \text{ thinlist}(Q) \sqsubseteq \text{thinlist}(Q)$ $\text{ordered}(\leq) \text{ — } \triangleq \text{thinlist}$
(c)	$\sqsubseteq \text{setify}^\circ \text{ thinlist ordered}(\leq)$	$\text{thinlist}(Q) \text{ setify} \sqsubseteq \text{setify thin}(Q) \text{ — } \triangleq$ $\text{thinlist, dom-laws, triple-chains}$
(d)	$\sqsubseteq \text{thin}(Q) \text{ setify}^\circ \text{ ordered}(\leq)$	$\text{sort}(\leq) \triangleq \text{setify}^\circ \text{ ordered}(\leq) \text{ — } \triangleq \text{thinlist}$
(e)	$= \text{thin}(Q) \text{ sort}(\leq)$	

Map(f) $\wedge$ ($\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)$) $\wedge$ MonoAlg(f, $\leq$) $\wedge$ sortF(f $\leq$ f $^\circ$) listf $\sqsubseteq$ P(f) sort($\leq$) $\wedge$ sort($\leq$) filterp $\sqsubseteq$ E(p) sort($\leq$) $\wedge$ F(sort($\leq$)) listcp $\sqsubseteq$ cp(F) sortF(F($\leq$)) $\Rightarrow$ F(sort($\leq$)) listcp listf filterp $\sqsubseteq$ $\frac{F(\exists) f p}{\exists}$ sort($\leq$) Lemma 8.1

one sorted list built from sorted arguments, instead of a set built and then sorted — f : FA $\rightarrow$ A monotonic on $\leq$, p coreflexive, F linear.

Map(f) $\wedge$ sort($\leq$) filterp $\sqsubseteq$ E(p) sort($\leq$) $\Rightarrow$ cp(F) P(f) sort($\leq$) filterp $\sqsubseteq$ $\frac{F(\exists) f p}{\exists}$ sort($\leq$)

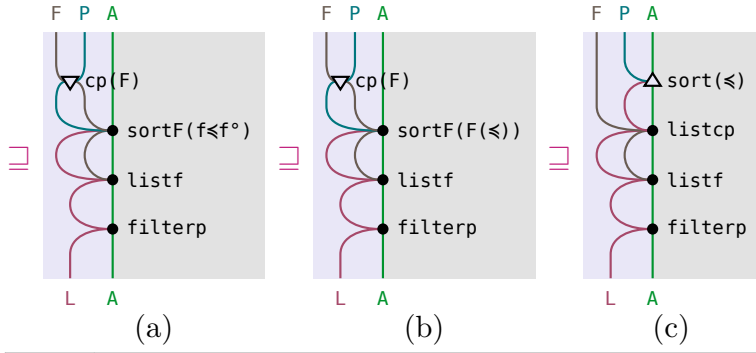
sorting the f-images of the product and filtering by p is below sorting the set F($\exists$)fp builds



(a)	$\frac{F(\exists) f p}{\exists} \text{ sort}(\leq)$	$\frac{F(\exists) fp}{\exists} = \frac{F(\exists)}{\exists} E(fp)$ — pow-laws, $\triangleq$ thinlist $E(f)=P(f)$, f a map — powrel-laws
(b)	$= \text{cp}(F) P(f) E(p) \text{ sort}(\leq)$	$\text{sort}(\leq) \text{ filter}(p) \sqsubseteq E(p) \text{ sort}(\leq)$ — (8.3b)
(c)	$\sqsubseteq \text{cp}(F) P(f) \text{ sort}(\leq) \text{ filterp}$	$\text{sort}(f \leq f^\circ) \text{ list}(f) \sqsubseteq P(f) \text{ sort}(\leq)$ — (8.3b)

Map(f) $\wedge$ ($\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)$) $\wedge$ MonoAlg(f, $\leq$) $\wedge$ sortF(f $\leq$ f $^\circ$) listf $\sqsubseteq$ P(f) sort($\leq$) $\wedge$ F(sort($\leq$)) listcp $\sqsubseteq$ cp(F) sortF(F($\leq$)) $\Rightarrow$ F(sort($\leq$)) listcp listf filterp $\sqsubseteq$ cp(F) P(f) sort($\leq$) filterp

combining the arguments' sorted lists by listcp, mapping f and filtering is below sorting the f-images and filtering



(a)	$\sqsubseteq \text{cp}(F) \text{ sortF}(f \leq f^\circ) \text{ listf filterp}$	$F(\leq) \sqsubseteq f \leq f^\circ$ — mon-str, $\triangleq$ thinlist
(b)	$\sqsubseteq \text{cp}(F) \text{ sortF}(F(\leq)) \text{ listf filterp}$	$F(\text{sort}(\leq)) \text{ listcp} \sqsubseteq \text{cp}(F) \text{ sort}(F(\leq))$ — (8.3b)
(c)	$\sqsubseteq F(\text{sort}(\leq)) \text{ listcp listf filterp}$	

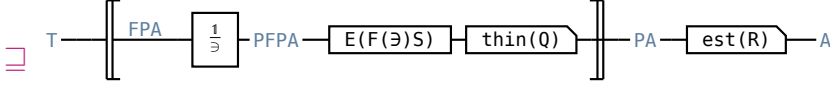
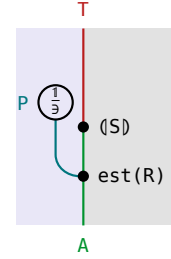
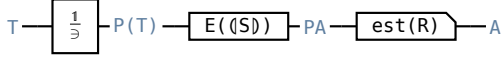
Theorem 8.2

$\text{Map}(f_1) \wedge \text{Map}(f_2) \wedge Q \sqsubseteq R \wedge 1 \sqsubseteq Q \wedge Q \sqsubseteq Q \wedge R^\circ \sqsubseteq R^\circ \wedge \text{MonoAlg}(f_1 \text{ } p_1, Q) \wedge$
 $\text{MonoAlg}(f_2 \text{ } p_2, Q) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \text{MonoAlg}(f_1, \leq) \wedge$
 $\text{MonoAlg}(f_2, \leq) \wedge \text{sortF}(f_1 \leq f_1^\circ) \text{ listf}_1 \sqsubseteq P(f_1) \text{ sort}(\leq) \wedge \text{sortF}(f_2 \leq f_2^\circ) \text{ listf}_2 \sqsubseteq$
 $P(f_2) \text{ sort}(\leq) \wedge \text{sort}(\leq) \text{ filterp}_1 \sqsubseteq E(p_1) \text{ sort}(\leq) \wedge \text{sort}(\leq) \text{ filterp}_2 \sqsubseteq E(p_2) \text{ sort}(\leq)$
 $\wedge F(\text{sort}(\leq)) \text{ listcp} \sqsubseteq \text{cp}(F) \text{ sortF}(F(\leq)) \wedge (\text{sort}(\leq) \times \text{sort}(\leq)) \text{ merge}(\leq) \sqsubseteq \text{cup} \text{ sort}(\leq)$
 $\wedge \text{sort}(\leq) \text{ thinlist}(Q) \sqsubseteq \text{thin}(Q) \text{ sort}(\leq) \wedge \text{sort}(\leq) \text{ minlist}(R) \sqsubseteq \text{est}(R) \wedge g_1 = \text{listf}_1$
 $\text{filterp}_1 \wedge g_2 = \text{listf}_2 \text{ filterp}_2 \wedge S = (f_1 \text{ } p_1) \cup (f_2 \text{ } p_2) \Rightarrow (\text{listcp} \langle g_1, g_2 \rangle \text{ merge}(\leq)$
 $\text{thinlist}(Q)) \text{ minlist}(R) \sqsubseteq \frac{(S)}{3} \text{ est}(R)$

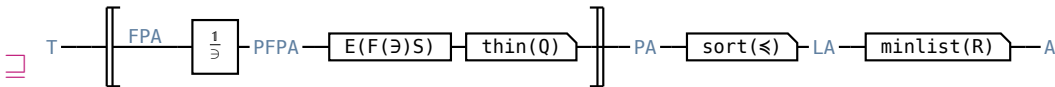
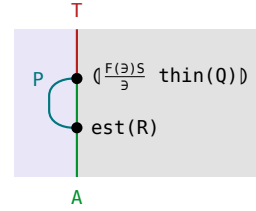
a fold on sorted lists of partial solutions, thinned at every step — at $\triangleq \text{thinlist}$'s binary thinning data.

circuit

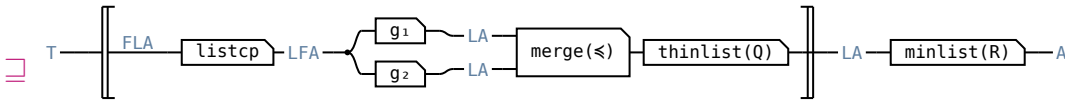
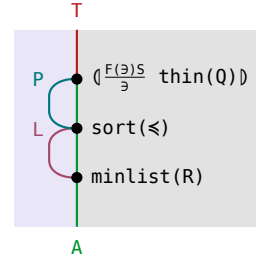
Hinze–Marsden



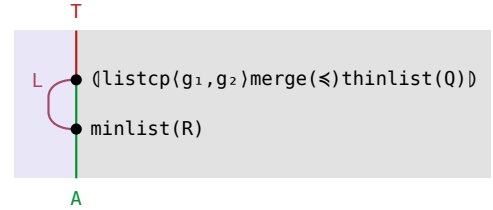
Corollary 8.1 at $f_1 p_1$ and $f_2 p_2$ monotonic on $Q \triangleq \text{thinlist}$



$\text{sort}(\leq) \text{ minlist}(R) \sqsubseteq \text{est}(R) \text{ — (8.3b) at its } Q \triangleq R$



fusion at (8.3c)



$$\begin{aligned}
& \text{Map}(f_1) \wedge \text{Map}(f_2) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \text{MonoAlg}(f_1, \leq) \wedge \text{MonoAlg}(f_2, \leq) \wedge \quad (8.3c) \\
& \text{sortF}(f_1 \leq f_1^\circ) \text{ listf}_1 \sqsubseteq P(f_1) \text{ sort}(\leq) \wedge \text{sortF}(f_2 \leq f_2^\circ) \text{ listf}_2 \sqsubseteq P(f_2) \text{ sort}(\leq) \wedge \text{sort}(\leq) \\
& \text{filterp}_1 \sqsubseteq E(p_1) \text{ sort}(\leq) \wedge \text{sort}(\leq) \text{ filterp}_2 \sqsubseteq E(p_2) \text{ sort}(\leq) \wedge F(\text{sort}(\leq)) \text{ listcp} \sqsubseteq \text{cp}(F) \\
& \text{sortF}(F(\leq)) \wedge (\text{sort}(\leq) \times \text{sort}(\leq)) \text{ merge}(\leq) \sqsubseteq \text{cup} \text{ sort}(\leq) \wedge \text{sort}(\leq) \text{ thinlist}(Q) \sqsubseteq \text{thin}(Q) \\
& \text{sort}(\leq) \Rightarrow F(\text{sort}(\leq)) \text{ listcp} (\text{listf}_1 \text{ filterp}_1, \text{listf}_2 \text{ filterp}_2) \text{ merge}(\leq) \text{ thinlist}(Q) \sqsubseteq \\
& \quad \frac{F(\exists) ((f_1 p_1) \cup (f_2 p_2))}{\exists} \text{ thin}(Q) \text{ sort}(\leq)
\end{aligned}$$

sorting the candidate set is what turns the thinning algebra into an algebra on lists — the side condition of binary thinning theorem's last step.

circuit	Hinze–Marsden
$ \begin{aligned} & \text{FPA} \xrightarrow{\frac{1}{3}} \text{PFPA} \xrightarrow{E(F(\exists)S)} \text{PA} \xrightarrow{\text{sort}(\leq)} \text{thinlist}(Q) \xrightarrow{\text{LA}} \\ & \text{sort}(\leq) \text{ thinlist}(Q) \sqsubseteq \text{thin}(Q) \text{ sort}(\leq) \quad (8.3b) \end{aligned} $	
$ \begin{aligned} & \text{FPA} \xrightarrow{\frac{1}{3}} \text{PFPA} \xrightarrow{E(F(\exists)f_1 p_1)} \text{PA} \xrightarrow{\text{cup}} \text{PA} \xrightarrow{\text{sort}(\leq)} \text{thinlist}(Q) \xrightarrow{\text{LA}} \\ & = \frac{F(\exists)f_1 p_1}{\exists}, \frac{F(\exists)f_2 p_2}{\exists} \text{ cup } \text{sort}(\leq) \text{ thinlist}(Q) \\ & S = (f_1 p_1) \cup (f_2 p_2) \quad \triangleq \text{thinlist}, \text{ then } \triangleq \text{cup} \end{aligned} $	
$ \begin{aligned} & \text{FPA} \xrightarrow{\frac{1}{3}} \text{PFPA} \xrightarrow{E(F(\exists)f_1 p_1)} \text{PA} \xrightarrow{\text{sort}(\leq)} \text{merge}(\leq) \xrightarrow{\text{thinlist}(Q)} \text{LA} \\ & = \frac{F(\exists)f_1 p_1}{\exists}, \frac{F(\exists)f_2 p_2}{\exists} (\text{sort}(\leq) \times \text{sort}(\leq)) \text{ merge}(\leq) \text{ thinlist}(Q) \\ & (\text{sort}(\leq) \times \text{sort}(\leq)) \text{ merge}(\leq) \sqsubseteq \text{cup } \text{sort}(\leq) \quad (8.3b) \end{aligned} $	
$ \begin{aligned} & \text{FPA} \xrightarrow{\frac{1}{3}} \text{PFPA} \xrightarrow{E(F(\exists)f_1 p_1)} \text{PA} \xrightarrow{\text{sort}(\leq)} \text{LA} \xrightarrow{\text{merge}(\leq)} \text{thinlist}(Q) \xrightarrow{\text{LA}} \\ & = \frac{F(\exists)f_1 p_1}{\exists} \text{ sort}(\leq), \frac{F(\exists)f_2 p_2}{\exists} \text{ sort}(\leq) \text{ merge}(\leq) \text{ thinlist}(Q) \\ & (X, Y) (\text{sort}(\leq) \times \text{sort}(\leq)) = (X \text{ sort}(\leq), Y \text{ sort}(\leq)) \quad \text{— bdm-prod-laws} \end{aligned} $	
$ \begin{aligned} & \text{FPA} \xrightarrow{F(\text{sort}(\leq))} \text{FLA} \xrightarrow{\text{listcp}} \text{LFA} \xrightarrow{\text{listf}_1 \text{ filterp}_1, \text{listf}_2 \text{ filterp}_2} \text{LA} \xrightarrow{\text{merge}(\leq)} \text{thinlist}(Q) \xrightarrow{\text{LA}} \\ & \text{F}(\text{sort}(\leq)) \text{ listcp} (g_1, g_2) \text{ merge}(\leq) \text{ thinlist}(Q) \\ & \text{Lemma 8.1 at } f_1, p_1 \text{ and at } f_2, p_2, \text{ then } X(g_1, g_2) \sqsubseteq (Xg_1, Xg_2) \quad \text{— bdm-prod-laws; } g_i \triangleq \text{list}(f_i) \\ & \text{filter}(p_i) \quad \triangleq \text{thinlist} \end{aligned} $	

§8.4. The knapsack problem

(8.4a)

`vol, wt : Item → Real, value ≐ list(vol) sum, weight ≐ list(wt) sum .`

`subseq = ([nil, cons] ∪ [nil, π2]) ∘ , ,` within `w` the coreflexive on `xs` with weight `xs ≤ w`, `0 ≤ w`.

`R ≐ value ≥ value°`, `Q ≐ Rn(weight ≤ weight°)`, `P ≐ R`, .

`FA = 1 + Item × A`, `listcp = wrap + cpr`, `g1 ≐ list([nil, cons])` `filter(within w)` `= [list(nil), h1]`,
`g2 ≐ list([nil, π2]) = [list(nil), h2]`.

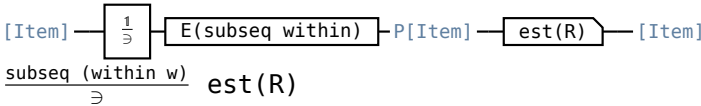
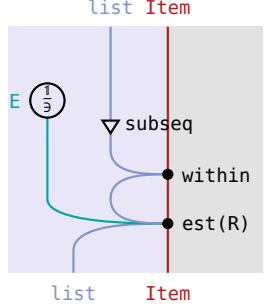
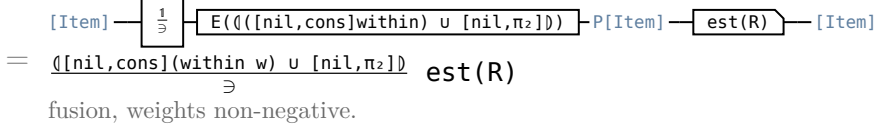
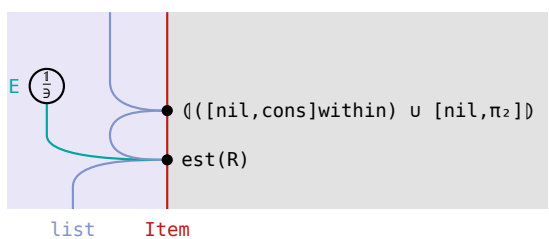
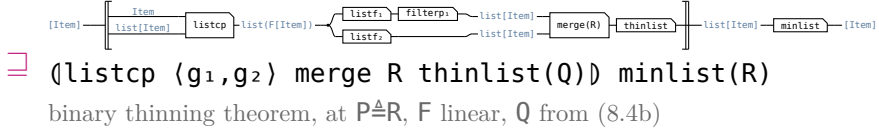
`h1 ≐ list(cons) filter(within w)`, `h2 ≐ list(π2)`.

(8.4b)

the law	what it says
$(1 \times R) \text{ (cons (within } w)) \sqsubseteq \text{cons (within } w) R$ FALSE	not monotonic on R: a selection of greater value need not still fit once one more item goes in
$(1 \times Q) \text{ (cons (within } w)) \sqsubseteq \text{cons (within } w) Q$ $(1 \times Q) \pi_2 \sqsubseteq \pi_2 Q$,	both halves are monotonic on Q once ties in value are broken by weight

$0 \leq w \wedge (\forall i. 0 \leq \text{wt}(i)) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \text{sortF}([\text{nil}, \text{cons}] R$
 $[\text{nil}, \text{cons}]^\circ) \text{ listf}_1 \sqsubseteq P([\text{nil}, \text{cons}]) \text{ sort}(R) \wedge \text{sortF}([\text{nil}, \pi_2] R [\text{nil}, \pi_2]^\circ) \text{ listf}_2 \sqsubseteq$
 $P([\text{nil}, \pi_2]) \text{ sort}(R) \wedge \text{sort}(R) \text{ filterp}_1 \sqsubseteq E(\text{within}) \text{ sort}(R) \wedge F(\text{sort}(R)) \text{ listcp} \sqsubseteq \text{cp}(F)$
 $\text{sortF}(F(R)) \wedge (\text{sort}(R) \times \text{sort}(R)) \text{ merge}(R) \sqsubseteq \text{cup sort}(R) \wedge \text{sort}(R) \text{ thinlist} \sqsubseteq \text{thin}(Q(\text{vol}, \text{wt}))$
 $\text{sort}(R) \wedge \text{sort}(R) \text{ minlist} \sqsubseteq \text{est}(R) \Rightarrow \langle \text{listcp} (\text{listf}_1 \text{ filterp}_1, \text{listf}_2) \text{ merge}(R) \text{ thinlist} \rangle$
 $\text{minlist} \sqsubseteq \frac{\text{subseq within}}{\supset} \text{est}(R)$

the knapsack problem, as a fold that thins the packings kept at each item

circuit	Hinze–Marsden
 subseq (within w) $\supset$ est(R)	
 fusion, weights non-negative.	
 binary thinning theorem, at $P^A R$, F linear, Q from (8.4b)	
$\langle [\text{nil}, \text{cpr} \langle h_1, h_2 \rangle \text{ merge } R \text{ thinlist}(Q)] \rangle \text{ minlist}(R)$ $= \text{listcp} = \text{wrap} + \text{cpr}, g_i = [\text{list}(\text{nil}), h_i] \text{ — } \triangleq \text{within};$ minlist(R) is head, packings coming out in descending value	

§8.5. The paragraph problem

(8.5a)

```

Line=list+ Word, Para=list+ Line, FA=Word+Word×A, listcp=wrap+cpr.
new(a,xs)=[[a]]#xs, glue(a,xs)=[[a]#head(xs)]#tail(xs), partition≐([wrap wrap,new u glue]) : list+
Word→Para.

width≐([length,(length×1) plus succ]), 0≤length a, fits w the coreflexive on a line x with width x≤w, ok w the
coreflexive on [x]#xs with width x≤w.

white w x=w-width x, collect≐list(sqr) sum, waste w≐init list(white w) collect.

R≐(waste w)≤(waste w)°, Q≐Rn(head head°), P≐T, .

g1≐list([wrap wrap,new]), g2≐list([wrap wrap,glue]) filter(ok w), start≐wrap wrap wrap, h1≐list(new),
h2≐list(glue) filter(ok w).

```

(8.5b)

the law	what it says
(1×R) glue⊆glue R FALSE	glue is not monotonic on R: the waste of a paragraph depends on its whole first line, so no greedy algorithm solves this
(1×Q) new⊆new Q (1×Q) (glue (ok w))⊆glue (ok w)Q cons monotonic on collect≤collect°.	both halves are monotonic on Q once ties in waste are broken by the first line
merge T=cat; P≐head prefix head° also serves	T needs no sorting at all, and prefix is a linear order on first lines of paragraphs of one input

($\forall a. 0 \leq \text{len}(a) \wedge (\forall a. \text{len}(a) \leq w) \wedge (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \text{sortF}([\text{wrap}$ (8.5c)
 $\text{wrap}, \text{new}] \sqsubseteq [\text{wrap wrap}, \text{new}]^\circ) \text{listf}_1 \sqsubseteq P([\text{wrap wrap}, \text{new}]) \text{sort}(\tau) \wedge \text{sortF}([\text{wrap wrap}, \text{glue}] \sqsubseteq$
 $[\text{wrap wrap}, \text{glue}]^\circ) \text{listf}_2 \sqsubseteq P([\text{wrap wrap}, \text{glue}]) \text{sort}(\tau) \wedge \text{sort}(\tau) \text{filterp}_2 \sqsubseteq E(\text{ok}) \text{sort}(\tau) \wedge$
 $F(\text{sort}(\tau)) \text{listcp} \sqsubseteq \text{cp}(F) \text{sortF}(F(\tau)) \wedge (\text{sort}(\tau) \times \text{sort}(\tau)) \text{merge}(\tau) \sqsubseteq \text{cup} \text{sort}(\tau) \wedge \text{sort}(\tau)$
 $\text{thinlist} \sqsubseteq \text{thin}(Q(\text{len}, w)) \text{sort}(\tau) \wedge \text{sort}(\tau) \text{minlist} \sqsubseteq \text{est}(R) \Rightarrow ([\text{listcp} (\text{listf}_1, \text{listf}_2$
 $\text{filterp}_2) \text{merge}(\tau) \text{thinlist}) \text{minlist} \sqsubseteq \frac{\text{partition fits}}{\supset} \text{est}(R)$
 a paragraph laid out as a fold that thins the layouts kept at each word

circuit	Hinze–Marsden
$\text{list}^*(\text{Word}) \xrightarrow{\frac{1}{2}} P(\text{list}^*(\text{Word})) \xrightarrow{E(\text{partition fits})} P(\text{list}^*(\text{list}^*(\text{Word}))) \xrightarrow{\text{est}(R)} \text{list}^*(\text{list}^*(\text{Word}))$ $\frac{\text{partition list}^*(\text{fits } w)}{\supset} \text{est}(R)$	
$\text{list}^*(\text{Word}) \xrightarrow{\frac{1}{2}} P(\text{list}^*(\text{Word})) \xrightarrow{E([\text{wrap wrap}, \text{new}] \sqcup ([\text{wrap wrap}, \text{glue}] \text{ok}]))} P(\text{list}^*(\text{list}^*(\text{Word}))) \xrightarrow{\text{est}(R)} \text{list}^*(\text{list}^*(\text{Word}))$ $= \frac{([\text{wrap wrap}, \text{new}] \sqcup (\text{glue } (\text{ok } w)))}{\supset} \text{est}(R)$ fusion, every word fits on a line by itself.	
$\text{list}^*(\text{Word}) \xrightarrow{\frac{1}{2}} P(\text{list}^*(\text{Word})) \xrightarrow{E([\text{wrap wrap}, \text{new}] \sqcup ([\text{wrap wrap}, \text{glue}] (\text{ok } w)))} P(\text{list}^*(\text{list}^*(\text{Word}))) \xrightarrow{\text{est}(R)} \text{list}^*(\text{list}^*(\text{Word}))$ $= \frac{([\text{wrap wrap}, \text{new}] \sqcup ([\text{wrap wrap}, \text{glue}] (\text{ok } w)))}{\supset} \text{est}(R)$ the algebra as $(f_1 p_1) \sqcup (f_2 p_2)$, $p_1 \triangleq 1 \triangleq \text{thinlist}$.	
$\text{list}^*(\text{Word}) \xrightarrow{\frac{1}{2}} P(\text{list}^*(\text{Word})) \xrightarrow{E([\text{wrap wrap}, \text{new}] \sqcup ([\text{wrap wrap}, \text{glue}] (\text{ok } w)))} P(\text{list}^*(\text{list}^*(\text{Word}))) \xrightarrow{\text{est}(R)} \text{list}^*(\text{list}^*(\text{Word}))$ $\sqsubseteq ([\text{listcp} (g_1, g_2) \text{cat thinlist}(Q)] \text{minlist}(R))$ binary thinning theorem, at $P \triangleq T$ with $\text{merge } \tau = \text{cat}$, Q from (8.5b)	
$([\text{start}, \text{cpr} (h_1, h_2) \text{cat thinlist}(Q)] \text{minlist}(R))$ $= \text{listcp} = \text{wrap} + \text{cpr}, g_i \text{ along the coproduct } \triangleq \text{partition}$	

§8.6. Bitonic tours

(8.6a)

$\text{FA} = (\text{City} \times \text{City}) + (\text{City} \times A)$, the base functor of cons-lists of length at least two; $\text{listcp} = \text{wrap} + \text{cpr}$; $\text{tc} : \text{City} \times \text{City} \rightarrow \text{Real}$, neither positive nor symmetric.

$\text{start } (a, b) = ([a, b], [a, b])$, $\text{dropl } (a, ([b] \# xs, ys)) = ([a] \# xs, [a] \# ys)$, $\text{dropr } (a, (xs, [b] \# ys)) = ([a] \# xs, [a] \# ys)$, $\text{tour} \triangleq ([\text{start}, \text{dropl } \sqcup \text{dropr}])$.

$\text{cost } (xs, ys) = \text{outcost } xs + \text{incost } ys$, $\text{outcost } [a_0, \dots, a_n] = \text{tc } (a_0, a_1) + \dots + \text{tc } (a_{n-1}, a_n)$, $\text{incost } [a_0, \dots, a_n] = \text{tc } (a_1, a_0) + \dots + \text{tc } (a_n, a_{n-1})$.

$\text{next} \triangleq \text{tail head}$, $\text{next2} \triangleq \text{next} \times \text{next}$, $\text{head2} \triangleq \text{head} \times \text{head}$, $R \triangleq \text{cost} \leq \text{cost}^\circ$, $Q \triangleq R \cap (\text{next2 next2}^\circ) \cap (\text{head2 head2}^\circ)$, $P \triangleq T$, $g_1 \triangleq \text{list}([\text{start}, \text{dropl}])$, $g_2 \triangleq \text{list}([\text{start}, \text{dropr}])$.

(8.6b)

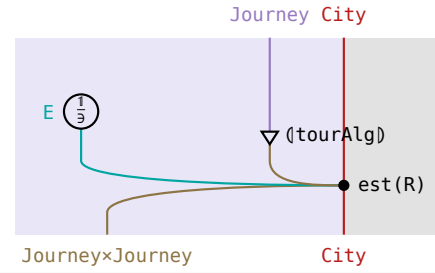
the law	what it says
$(1 \times R) \text{dropl} \sqsubseteq \text{dropl } R$ FALSE	neither drop is monotonic on R: the two edges it adds and removes depend on head and next of both lists
$(1 \times R) \text{dropr} \sqsubseteq \text{dropr } R$ FALSE	
$(1 \times Q) \text{dropl} \sqsubseteq \text{dropl } Q$	both are, once ties in cost are broken by the two second cities — the heads already agree among tours of one input
$(1 \times Q) \text{dropr} \sqsubseteq \text{dropr } Q$,	

$$\begin{aligned}
& (\forall X. \forall Y. X \sqsubseteq Y \Rightarrow \text{sortF}(X) \sqsubseteq \text{sortF}(Y)) \wedge \text{sortF}([\text{start}, \text{droplFn}] \tau [\text{start}, \text{droplFn}]^\circ) g_1 \sqsubseteq \quad (8.6c) \\
& P([\text{start}, \text{droplFn}]) \text{sort}(\tau) \wedge \text{sortF}([\text{start}, \text{droprFn}] \tau [\text{start}, \text{droprFn}]^\circ) g_2 \sqsubseteq \\
& P([\text{start}, \text{droprFn}]) \text{sort}(\tau) \wedge F(\text{sort}(\tau)) \text{listcp} \sqsubseteq \text{cp}(F) \text{sortF}(F(\tau)) \wedge (\text{sort}(\tau) \times \text{sort}(\tau)) \text{cat} \\
& \sqsubseteq \text{cup} \text{sort}(\tau) \wedge \text{sort}(\tau) \text{thinlist} \sqsubseteq \text{thin}(Qc(tc)) \text{sort}(\tau) \wedge \text{sort}(\tau) \text{minlist} \sqsubseteq \text{est}(R) \Rightarrow (\text{listcp} \\
& (g_1, g_2) \text{cat thinlist}) \text{minlist} \sqsubseteq \frac{(\text{tourAlg})}{\exists} \text{est}(R) \\
& \text{a least-cost bitonic tour, as a fold that thins the tours kept at each city}
\end{aligned}$$

circuit

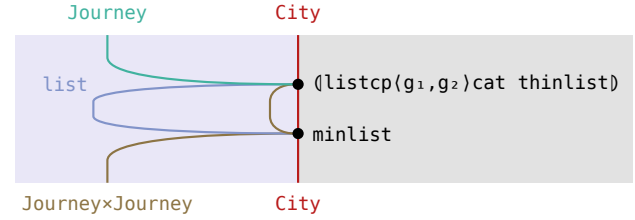
Hinze–Marsden

$$\begin{aligned}
& \text{dCL}(\text{City} \times \text{City}, \text{City}) \xrightarrow{\frac{1}{\exists}} P(\text{dCL}(\text{City} \times \text{City}, \text{City})) \xrightarrow{E(\text{tourAlg})} P(\text{Journey}^2) \xrightarrow{\text{est}(R)} \text{Journey} \\
& \frac{\text{tour}}{\exists} \text{est}(R)
\end{aligned}$$



$$\begin{aligned}
& \text{dCL}(\text{City} \times \text{City}, \text{City}) \xrightarrow{\frac{1}{\exists}} P(\text{dCL}(\text{City} \times \text{City}, \text{City})) \xrightarrow{E([\text{start}, \text{droplFn}] \cup [\text{start}, \text{droprFn}])} P(\text{Journey}^2) \xrightarrow{\text{est}(R)} \text{Journey} \\
& = \frac{([\text{start}, \text{dropl}] \cup \text{dropr})}{\exists} \text{est}(R) \\
& \text{tour} \triangleq ([\text{start}, \text{dropl}] \cup \text{dropr}) \triangleq \text{tour}
\end{aligned}$$

$$\begin{aligned}
& \text{dCL}(\text{City} \times \text{City}, \text{City}) \xrightarrow{F(\text{list}(\text{Journey}^2))} \text{listcp} \xrightarrow{\text{list}(F(\text{Journey}^2))} \begin{matrix} g_1 \\ g_2 \end{matrix} \xrightarrow{\text{cat}} \text{thinlist} \xrightarrow{\text{list}(\text{Journey}^2)} \text{minlist} \xrightarrow{\text{Journey}} \text{Journey} \\
& \sqsubseteq (\text{listcp} (g_1, g_2) \text{cat thinlist}(Q)) \text{minlist}(R) \\
& \text{binary thinning theorem, at } P \triangleq T \text{ with merge } \tau = \text{cat}, Q \text{ from (8.6b)}
\end{aligned}$$



$$\begin{aligned}
& ([\text{start wrap}, \text{cpr} (\text{list}(\text{dropl}), \text{list}(\text{dropr})) \text{cat} \\
& = \text{thinlist}(Q)] \text{minlist}(R) \\
& \text{listcp} = \text{wrap} + \text{cpr}, g_i = [\text{list}(\text{start}), \text{list}(\text{drop}_i)] \triangleq \text{tour}; \\
& \text{quadratic, two tours added per step}
\end{aligned}$$